NATIONAL CANCER INSTITUTE

What Are Clinical Trials?

Clinical trials are research studies that test how well new medical approaches work in people.

Talk with your doctor about clinical trials when you discuss treatment options.
Credit: SDI Productions

What is the purpose of a clinical trial?

Clinical trials test new ways to find, prevent, and treat cancer. They also help doctors improve the quality of life for people with cancer by testing ways to manage the side effects of cancer and its treatment.

Why are clinical trials important?

Today, people are living longer lives thanks to results of past cancer clinical trials. When you take part in a clinical trial, you add to our knowledge about cancer and help improve cancer care for people in the future.

People join clinical trials for many reasons. People living with cancer often join trials because they want to help future patients. People with certain risk factors want to help doctors learn how to prevent cancer. Healthy volunteers want to help doctors learn how to find cancer early.

People with cancer and healthy volunteers join trials to play a role in cancer research and move science forward to help others.

What are the types of clinical trials?

There are several types of cancer clinical trials. Each type of trial is designed to answer different research questions and will help researchers learn things that can help people in the future.

Treatment Trials

Most cancer clinical trials are treatment studies that involve people who have cancer. These trials test new treatments or new ways of using existing treatments, including new:

- drugs
- vaccines
- approaches to surgery or radiation therapy
- combinations of treatments

"I feel that as an African American we do not participate in programs that could BE and sometimes ARE beneficial to us. I hope by my participation this will encourage others African American women to be more open." —Marsha Dukes, NCI clinical trial participant

As researchers learn more about the genetic changes that lead to cancer, doctors are testing treatments that target these specific changes. So in some treatment trials, your tumor may be tested to see if treatments targeting specific genetic changes might work better than standard treatments.

Treatment trials are designed to answers questions such as:

- What is a safe dose of the treatment under study?
- How should the treatment be given?
- Does the treatment help people with cancer live longer than current treatment?
- Can the study treatment shrink tumors or slow their growth and spread?
- What are the treatment's side effects?
- Does the study treatment allow for a better quality of life with fewer side effects?
- Does the treatment help delay the return of the cancer?

Prevention Trials

Prevention trials are studies that look at ways to prevent cancer.

In most prevention trials, the people who take part do not have cancer but are at high risk for developing it. Or they

Participate in Cancer Prevention Studies

Learn about cancer prevention

have had cancer and are at high risk for developing a new cancer.

studies and how to participate.

There are two kinds of prevention trials, action studies and agent studies. In action studies, you are asked to do something, such as exercise or follow a special diet. In agent studies, you are asked to take something, such as a drug or vitamin. Learn about participating in prevention studies.

Researchers who conduct these studies want to know:

- How safe is the drug or activity?
- Does the new approach reduce the chance that someone will get cancer?

Screening Trials

Cancer Screening Studies

Learn about joining a screening trial to help find cancer early.

The goal of cancer screening trials is to test ways to find cancer before it causes symptoms, when it may be easier to treat.

An effective screening test will reduce the number of people who die from the cancer that is being screened for. Learn about joining a cancer screening study.

But screening tests can have harms, which include bleeding or other physical damage. Other possible harms include a result that shows you might have cancer when you don't. When this happens, it may lead to unnecessary tests and procedures. On the other hand, the results may show no signs of cancer when you have it. And sometimes screening can find cancers that would not have harmed you during your lifetime.

Researchers who conduct cancer screening studies want to know:

- Does finding disease earlier, before people have any symptoms, save lives?
- Is one screening test better than another?
- Do the benefits of the screening test outweigh the harms?

Supportive Care/Palliative Care Trials

These trials look at ways to improve the quality of life of people with cancer, especially those who have side effects from cancer and its treatment.

They might test drugs, such as those that help with depression or nausea. Or they might test activities, such as attending support groups, exercising, or talking with a counselor.

Some of these trials test ways to help families and caregivers cope with their own needs, as well as those of the person with cancer.

Researchers who conduct these studies want to know:

- How does cancer and its treatment affect patients and their loved ones?
- What can improve the comfort and quality of life of people who have cancer?

Related Resources

Find a Clinical Trial

How Do Clinical Trials Work?

Who Pays for Clinical Trials?

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